Visual Scene Understanding through Neuro-Symbolic Reasoning

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Abstract. We implement an interpretable neuro-symbolic visual question answering (VQA) system that reconstructs a scene graph from an image and executes a learned question program over it. Hybrid detection and segmentation, relative depth, and zero-shot attribute classification are combined with a fine-tuned T5-small semantic parser and a typed, deterministic executor. Evaluation separates program reasoning from visual perception: answer accuracy is **99.9%** with reference scene graphs versus **79.0%** end-to-end on CLEVR. Without target-domain program labels, REINFORCE improves accuracy on human-written questions by **15.7 percentage points**. Component-wise evaluation identifies color recognition and spatial relations as major bottlenecks.

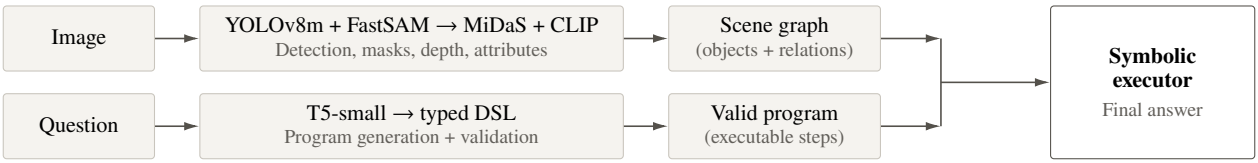


Figure 1. Explicit image and language pipelines joined by deterministic execution.

1. Problem and System Design

End-to-end VQA often obscures whether an error comes from scene perception or reasoning. Our system exposes inspectable *objects and relations*, *question-conditioned DSL programs*, and *execution traces*. Following NS-VQA [2], it combines modular visual processing with explicit program validation and deterministic reasoning.

2. Perception: Image to Scene Graph

Hybrid detection. A COCO-pretrained YOLOv8m detector was fine-tuned for **15 epochs** on **35,000 CLEVR images** with one generic object class. Low-threshold YOLO detections are matched to class-agnostic FastSAM proposals using box IoU (matching threshold 0.3); high-confidence unmatched proposals recover missed objects. Accepted boxes undergo crop-wise FastSAM segmentation followed by morphological opening and closing.

Depth and attributes. MiDaS_small supplies relative inverse depth. An occupancy-based, depth-ordered mask assignment resolves overlapping object masks. Zero-shot CLIP [5] (ViT-B/32) predicts color, shape, and material using averaged embeddings of three prompts per candidate attribute; the relative bounding-box area gives small/large size. Object centers and depth differences define *left/right/front/behind* relations. The output is an explicit JSON scene graph with object attributes, coordinates, and indexed relation lists.

3. Language to Executable Reasoning

Typed program representation. T5-small [4] maps a question to a linearized DSL program based on CLEVR's functional representation [1]. The implemented type system distinguishes *ObjectSet*, *Integer*, *Bool*, and *Attribute*; operators implement filtering, relations, counting, set operations, comparisons, and attribute queries. Beam search ($k = 5$) produces candidate programs, checked for structural validity, input types, and allowed arguments before execution.

Deterministic executor. Each operation consumes prior intermediate results. For example, a query such as "What color is the cube behind the cyan sphere?"" becomes `scene` → `filter_color[cyan]` → `filter_shape[sphere]` → `unique` → `relate[behind]` → `filter_shape[cube]` → `unique` → `query_color`. Invalid or ambiguous `unique` selections raise execution errors rather than guessing. Formally,

$$\hat{a} = \text{Execute}\big(P_\theta(q),\, G_\phi(I)\big)\,.$$

4. Training and Evaluation

Supervised parsing. T5-small was trained for **3 epochs** on CLEVR question–program pairs with token-level cross-entropy and AdamW ($5 \cdot 10^{-4}$ learning rate). **Reward-based adaptation.** CLEVR-Humans contains no target-domain program annotations. KL-regularized REINFORCE [6] ($\beta = 0.05$) samples **four** candidate programs per question and uses execution feedback: +1 correct answer, +0.2 executable but incorrect, −0.1 execution failure, and 0 invalid syntax.

Two evaluation conditions. *GT* means predicted programs executed on annotated scene graphs, isolating language and reasoning; *E2E* means the same logic applied to graphs reconstructed from images. CLEVR GT uses **1,000 questions**, whereas its E2E test uses **10,000 questions / 1,000 images**. CoGenT valB evaluates unseen attribute combinations (5,000 GT questions), CLEVR-Humans evaluates natural-language variation (7,202 validation questions), and perception is tested on 1,000 images.

5. Main Findings

Table 1. Answer accuracy (%), using reference (GT) or reconstructed (E2E) scene graphs.

Validation split	GT	E2E
CLEVR	**99.9**	79.0
CLEVR-CoGenT B	99.7	**79.0**
CLEVR-Humans, before RL	41.4	32.7
CLEVR-Humans, after RL	**57.1**	**45.6**

Reliable compositional reasoning. With reference scene graphs, CLEVR achieves **99.9%** accuracy, **100%** valid programs, and **0%** execution failures. CoGenT B remains at **99.7%**, indicating robust recombination of familiar attributes. However, reconstructing the scene graph reduces full-system accuracy to **79.0%** on both sets. Since GT and E2E CLEVR evaluations use different subset sizes, this contrast should be read as a diagnostic, not a paired per-question ablation.

Natural language transfer. On CLEVR-Humans, REINFORCE increases GT accuracy **41.4 → 57.1%** (+15.7 points), while E2E rises **32.7 → 45.6%** (+12.9 points). The execution-error rate falls **25.5 → 0.6%**, despite the absence of target-domain DSL annotations.

6. Where Does It Fail?

Table 2. Component-level perception results on 1,000 CLEVR images.

Component	Measured performance
Detection	Recall 97.5%; precision 88.7%; F1 92.9%; exact object count 53.0%
Attributes	Shape 95.5%; material 94.4%; size 92.8%; **color 87.8%**
Relations	Left 66.5%; right 66.7%; front 68.6%; **behind 44.3%**

Perceptual error propagation. Although the object recall is high, the model generates **0.80 false positives per image** on average; only 53.0% of images have the exact object count. Color is the least accurate attribute, with *brown* at **71%**. Multiplying independently measured color/shape/material accuracies gives $0.878 \cdot 0.955 \cdot 0.944 \approx \mathbf{79.2}\%$, close to the 79.0% E2E score; this is a heuristic illustration, *not* an estimated probability of full-system success. Depth-dependent *behind* relations reach only 44.3%.

Language coverage. 813 of 7,202 CLEVR-Humans validation questions (11.3%) are not expressible with the current DSL, including *between*, superlatives, and most-frequent-attribute queries. Lexical variation (*ball/sphere*, *box/cube*) additionally reduces accuracy on otherwise expressible questions.

7. Interpretation and Outlook

The strongest result is the separation of *symbolic reasoning reliability* from *perception quality*. Under published but not necessarily identical evaluation configurations, CLEVR E2E (79.0%) trails IEP (96.9%) and NS-VQA (99.8%); on CoGenT B it exceeds their reported 73.7% and 63.9% [2,3]. The architecture supports targeted improvements: a CLEVR-aware color classifier, stronger depth/relationship estimation, and a DSL expanded to human-language constructions.

Selected references

[1] J. Johnson et al. "CLEVR: A Diagnostic Dataset for Compositional Language and Elementary Visual Reasoning." *CVPR*, 2017.
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[6] R. J. Williams. "Simple Statistical Gradient-Following Algorithms for Connectionist Reinforcement Learning." *Machine Learning*, 1992.